ScamSentry AI

Backend Database Schema & Data Architecture Document

1. Purpose

This document defines the recommended backend data schema for the ScamSentry AI hackathon MVP. It supports anonymous scam reporting, identifier lookup, trusted evidence aggregation, area-level activity maps, anomaly alerts and dynamic licence state.

2. Database Architecture

Recommended MVP database: SQLite with SQLAlchemy ORM. The relational design can later be migrated to PostgreSQL with minimal model changes.

```
IDENTIFIERS ■ 1 ■■■■■■■< SCAM_REPORTS >■■■■■■■ 1 AREAS ■ ■■■■■■■> internal abuse-control signals
AREAS ■■■■■■■< ALERTS LICENCE_STATE (independent system state)
```

3. Core Table: identifiers

Stores normalized entities that users can look up or report. One identifier may have many scam reports.

Column	Type	Constraints / Description
id	INTEGER	Primary Key, auto-increment
identifier_type	TEXT	NOT NULL; phone | upi | url
raw_value	TEXT	Original submitted value where required for audit/debugging
normalized_value	TEXT	NOT NULL; normalized lookup value
created_at	DATETIME	NOT NULL
updated_at	DATETIME	NOT NULL

Recommended constraint: UNIQUE(identifier_type, normalized_value) for fast duplicate detection and lookup.

4. Core Table: areas

Stores approximate geographic areas used for privacy-preserving aggregation. Exact reporter coordinates are not exposed or required for the public MVP.

Column	Type	Constraints / Description
id	INTEGER	Primary Key
name	TEXT	NOT NULL; locality/city/area name
city	TEXT	City name
state	TEXT	State name
pincode	TEXT	Optional indexed pincode
latitude	REAL	Approximate area coordinate
longitude	REAL	Approximate area coordinate
created_at	DATETIME	NOT NULL

5. Core Table: scam_reports

Primary evidence table. A report is not automatically trusted; only reports with accepted status contribute to public risk scoring, maps and anomaly detection.

Column	Type	Constraints / Description
id	INTEGER	Primary Key
identifier_id	INTEGER	Foreign Key → identifiers.id
area_id	INTEGER	Foreign Key → areas.id
scam_type	TEXT	NOT NULL; controlled scam category
description	TEXT	Optional reporter details
status	TEXT	accepted | flagged | duplicate | rate_limited | rejected
trust_score	INTEGER	Internal 0–100 confidence/evidence score
reporter_fingerprint_hash	TEXT	Hashed internal abuse-control signal
source_ip_hash	TEXT	Optional hashed network-context signal
created_at	DATETIME	NOT NULL
updated_at	DATETIME	NOT NULL

Suggested categories: digital_arrest, upi_fraud, fake_kyc, phishing, fake_job, loan_scam, investment_fraud, fake_customer_care, otp_fraud, other.

Recommended indexes: INDEX(identifier_id, status), INDEX(area_id, created_at), INDEX(scam_type, created_at), INDEX(status, created_at).

6. Internal Reporter Abuse-Control Data

The MVP should preserve anonymity while still detecting report-bombing. Raw identities should not be exposed through APIs.

Column	Type	Purpose
id	INTEGER	Primary Key
fingerprint_hash	TEXT	Hashed anonymous reporter signal
window_start	DATETIME	Rate-limit window
report_count	INTEGER	Submissions in current window
last_report_at	DATETIME	Latest submission timestamp

7. Table: alerts

Stores unusual activity detected from accepted reports only.

Column	Type	Constraints / Description
id	INTEGER	Primary Key
area_id	INTEGER	Foreign Key → areas.id
alert_type	TEXT	spike | new_type

Column	Type	Constraints / Description
severity	TEXT	low | medium | high
scam_type	TEXT	Relevant category where applicable
current_count	INTEGER	Current observation count
baseline_count	REAL	Historical comparison baseline
anomaly_score	REAL	Calculated anomaly score
message	TEXT	Human-readable explanation
is_active	BOOLEAN	Whether alert is currently shown
created_at	DATETIME	NOT NULL

8. Table: licence_state

Maintains dynamic licence state. Updating a valid licence should restore access without a server restart.

Column	Type	Constraints / Description
id	INTEGER	Primary Key; normally one active row
license_key_hash	TEXT	Hashed licence value or verification reference
status	TEXT	active | expired | invalid
issued_at	DATETIME	Optional issue timestamp
expires_at	DATETIME	Licence expiry timestamp
updated_at	DATETIME	NOT NULL

9. Relationships

Relationship	Cardinality	Meaning
Identifier → ScamReport	1 : Many	One phone/UPI/URL can have multiple reports.
Area → ScamReport	1 : Many	One area can contain many reports.
Area → Alert	1 : Many	One area can have multiple anomaly alerts.

10. Backend Processing Flow

1. Receive API request.
2. Validate request using Pydantic schemas.
3. Check dynamic licence state for protected endpoints.
4. Classify and normalize phone/UPI/URL input.
5. Apply rate-limit and duplicate/report-bombing checks.
6. Create or retrieve the identifier record.
7. Store report with the appropriate trust status.
8. Use accepted reports for lookup aggregation and map calculations.
9. Run anomaly logic for area/category changes.
10. Create or update alerts when thresholds are satisfied.

11. Public Data vs Private/Internal Data

Public API May Return	Must Remain Internal / Private
Risk verdict and risk score	Raw IP address
Accepted report counts	Reporter fingerprint values
Scam categories	Exact reporter identity
Area-level activity	Exact reporter location
Alert summaries	Internal abuse-control signals

12. SQLAlchemy Model Implementation Order

1. Database connection and Base model
2. Area model
3. Identifier model
4. ScamReport model
5. Alert model
6. ReporterRateLimit / abuse-control model
7. LicenceState model

13. MVP Build Priority

• Priority 1: identifiers + areas + scam_reports

• Priority 2: POST /reports and GET /lookup

• Priority 3: anti-abuse and rate-limit logic

• Priority 4: map aggregation

• Priority 5: alerts/anomaly detection

• Priority 6: dynamic licence state

Recommended principle: keep the schema simple enough to build within the hackathon timeframe, but ensure that only trusted/accepted evidence affects public risk, map and alert results.